

Day 2 — Lab Record

Expected instrument readings & key notes

First circuit, variable resistance, voltage dividers (no screenshots were taken — these are the expected views)

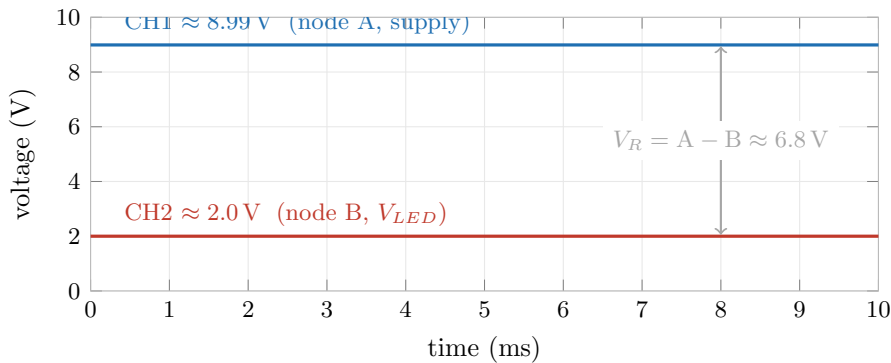
Which instrument each lab used:

- **Lab 1 (KVL)** and **Lab 4 (pot → LED)** were observed on the **oscilloscope** — so each has an expected *scope trace* below.
- **Lab 2 (Ohm's law sweep)** and **Lab 3 (divider loading)** were measured with the **DMM** — a voltage scope would not show current directly, and the DC voltages would simply appear as flat lines, so these are shown as the graph the readings actually produce.

Lab 1 — KVL on 9 V + 470 Ω + red LED

(scope, DC coupled)

Nodes: **A** = supply (+), **B** = junction between 470 Ω and LED, **C** = LED cathode = supply − = ground. **Both probe ground clips at C**. CH1 tip at A, CH2 tip at B. CH1 tip at A, CH2 tip at B.

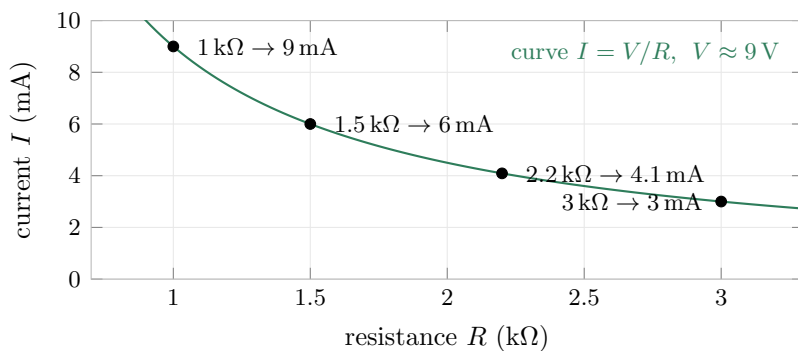


- A DC circuit shows as **flat horizontal lines** on the scope — nothing varies with time.
- The scope's value here is reading **two nodes at once**; the **Math channel** ($A - B$) gives V_R directly.
- **KVL check:** $V_R + V_{LED} = 6.8 + 2.0 = 8.8\text{ V} \approx$ the measured supply 8.99 V. ✓

Lab 2 — Ohm's law from a current sweep

(DMM in series, LED removed)

Just a resistor across the $\approx 9\text{ V}$ supply; DMM in **mA** mode, **in series**, so all current is read. $I = V/R$.

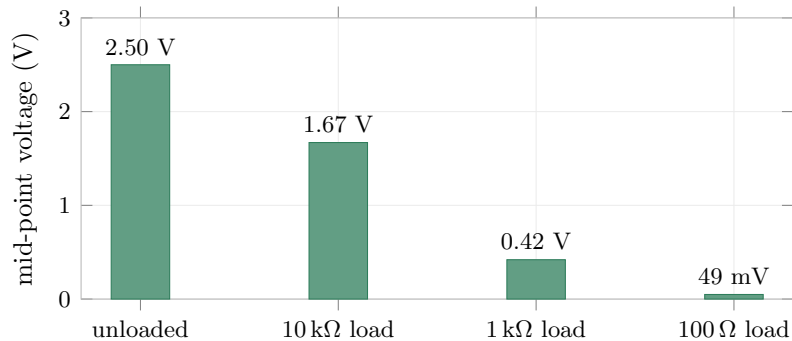


- Points sit on the $I = V/R$ curve, so $V = I \cdot R \approx 9\text{ V}$ at every value — that constancy *is* Ohm's law confirmed.
- Halving doesn't apply here, but note current **rises as R falls** (a brighter-LED intuition from earlier).

Lab 3 — Divider loading

(DMM at the mid-point)

10 k top, 10 k bottom, 5 V supply. Unloaded mid-point = 2.50 V, so $V_{Th} = 2.5$ V and $R_{Th} = 10\text{k} \parallel 10\text{k} = 5\text{k}\Omega$. Each load forms a divider with R_{Th} : $V = V_{Th} \frac{R_L}{R_{Th} + R_L}$.

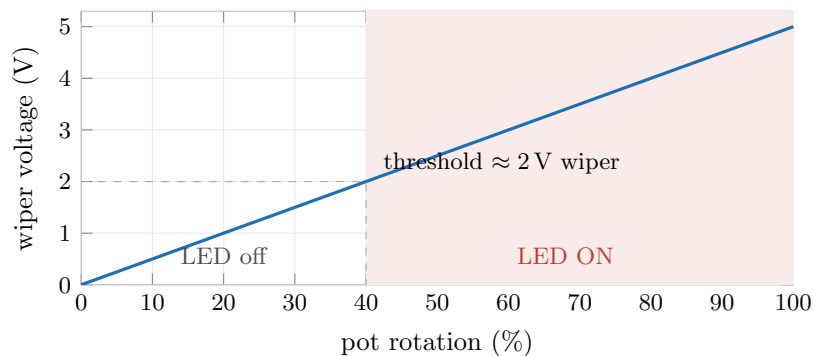


- The output is **not** a stiff 2.5 V — it **drops** as the load shrinks. This is the Thévenin story made visible.
- Smaller load $\Rightarrow$ more current through R_{Th} $\Rightarrow$ bigger drop across it. (Full working is in the EOD write-up.)
- On a scope each of these would simply be a **flat DC line** at that voltage.

Lab 4 — Pot-controlled LED

(scope on the wiper, DC coupled)

10 k pot ends to 5 V / GND; wiper $\rightarrow 470\Omega \rightarrow \text{LED} \rightarrow \text{GND}$. Turning the pot sweeps the wiper voltage; the curve below is wiper voltage vs rotation. On the scope this appears as a **flat line at the wiper voltage** that slides up or down as you turn.



- Below ~ 2 V at the wiper the LED is **off** (its ≈ 2 V forward drop isn't met); above it, brightness rises with wiper voltage.
- The 470Ω sets the current: $I = (V_{wiper} - 2)/470$, reaching $\approx 6.4\text{mA}$ at a 5 V wiper.

Key conceptual notes (worth keeping)

- **LED:** clamps its forward voltage at $\approx 2\text{ V}$ but does *not* limit current; **current (heat) is what destroys it**, so the series resistor's real job is to **limit current** — $I = (V - 2)/470$. You can't use $V/(R + R_{LED})$ because the LED is nonlinear; subtract the $\approx 2\text{ V}$ and use the resistor.
- **Potentiometer:** one resistive track *split by the wiper* into $R_{top} + R_{bot}$ (always summing to 10 k). The wiper is a **node, not a component**; its voltage is the divider output. The $470\ \Omega + \text{LED}$ is a **separate branch** hanging off the wiper node.
- **Parallel vs series (the key one):** parallel branches **always share the same voltage**; any voltage “**drop**”/**droop** is a *series* effect across R_{top} . With $R_{top} = 0$ there is no series resistance to drop voltage, so the wiper sits at the **full 5 V** and both parallel branches see 5 V at different currents.
- **Scope grounding:** all probe ground clips are internally **common and earth-tied** $\Rightarrow$ **tie every ground clip to one circuit-ground node**, or two clips will short two points together (this had shorted the LED earlier).
- **Thévenin / loading:** any linear two-terminal network looks like V_{Th} in series with R_{Th} ; a load forms a divider with R_{Th} , so the output droops. (Recipe and worked example in the EOD write-up.)